UNIVERSITY OF MINES AND TECHNOLOGY (UMaT), TARKWA

SCHOOL OF RAILWAYS AND INFRASTRUCTURE DEVELOPMENT
ESSIKADO CAMPUS

GEOMATIC ENGINEERING PROGRAMME

A PROJECT ENTITLED

MODELLING AND PREDICTION OF CUSTOMER LOYALTY OF LAND ADMINISTRATION SERVICE IN GHANA

BY:
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PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF DEGREE OF BACHELOR OF SCIENCE IN GEOMATIC ENGINEERING

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……..
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SEPTEMBER, 2025

DECLARATION

I hereby declare that this project is my original work. It is being submitted for the Bachelor of Science in Geomatic Engineering at the University of Mines and Technology School of Railways and Infrastructure Development (UMaT SRID), Essikado. It has not been submitted for any degree or examination in any other University.

…… . ……... day of ……, 2025.

(Signature of candidate

ABSTRACT

The goal of the study was to ascertain client loyalty. Numerous factors, including customer orientation and reception, customer happiness, service quality, transparency, and calm surroundings, are among them. Ten of the twelve observable variables were employed in total for the investigation. Confirmatory factor analysis and other statistical indicators were primarily employed to evaluate the model's performance and fitness. With the help of artificial neural networks (ANN), support vector machines (SVM), and structural equation models (SEM), this study attempts to forecast client loyalty. Using SEM, the relevant variables that influence loyalty were evaluated using statistical indicators. SVM and ANN (LSTM) modeling used the relevant variables derived from SEM modeling as input. Both models were chosen Based on their predicted accuracy and proficiency with non-linear models. The prediction error was less significant with MAPE, R^2, MAE, RMSE, and MSE values of 0.944, 0.013, 0.152, and 0.023, respectively for the SVM. The LSTM model had MAE, R^2, and RMSE values of 0.4904, 0.6145, and 0.5105.Customer loyalty was modeled and predicted by both LSTM and SVM, indicating that the combination of SEM-ANN and SEM-SVM works well as a functional model for identifying non-linear correlations between observed and unobserved factors.

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DEDICATION

I dedicate this project to my mentors, family members and friends whom with their efforts have helped me throughout my academic life. I am always grateful and thankful for your countless advices, inspirations and you’re strong and unwavering supports.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to God and the following individuals for their support and contributions to this project. My supervisor, Mr Felix Tabase, and the entire staff in the Geomatic engineering department of the School of Railways and Infrastructure Development (SRID) for their guidance, expertise, and encouragement throughout this project.

I am also thankful to my colleagues and peers, for their valuable feedback, suggestions, and support.

I am also grateful to my family and friends, for their unwavering support, patience, and understanding throughout this project.

I would also like to acknowledge the contributions of Mr Kenneth Appau whose work has been instrumental in shaping this project.

Thank you all for your help and support."

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CHAPTER 1
INTRODUCTION

1.1 Statement of the problem

Land Administration System (LAS) in Ghana are predominantly offered by the Lands Commission via a one-stop-shop called the Client Satisfaction Access Unit (CSAU). The CSAU is designed to aid transparency in land transactions, reduce corruption and the turnaround time in land services. This land services include land title and deeds registration, conveyancing, assignments, land valuation, official search, cadastral surveying and mapping and many more.

However, customers of the CSAU have complained about longer turnaround time, unnecessary bureaucratic process, corruption and other vices in the land administration services. Such practices has thereby result in high land dispute cases, defrauding of prospective interest holders to land, and a higher cost of land administration services (Kauffman *et al,* 2020). Therefore the need for the loyalty of clients to the CSAU is to be assessed, modelled and predicted. Despite the need for measurement, modelling and prediction of Customer Loyalty (CL), there is the lack of works done for such a purpose in the LAS.

This study therefore seeks to model and assess CL of the CSAU using the Structural Equation Modelling (SEM) and then predict CL using Artificial Intelligence (AI) methods such as Artificial Neural Networks (ANN) and Support Vector Machine (SVM).

1.2 Objectives of the Project

The objectives of the project are:

i. Model and assess CL using SEM;
ii. Predict CL using AI models including ANN and SVM of the CSAU of Ghana.
iii. Assessment and analysis of the model

1.3 Expected Outcomes

At the end of the project, it is expected that**:**

i. Customer loyalty of the CSAU of Ghana shall be modelled and assessed using SEM; and
ii. Customer Loyalty of the CSAU of Ghana shall be predicted using AI techniques including ANN and SVM.

1.4 Methods Used

The methods to be used are:

i. Development and dissemination of questionnaires using the Kobotoolbox;
ii. SEM will be used in modelling and assessment of key loyalty factors;
iii. Artificial Neural Network (ANN) and Support Vector Machine (SVM) will be used in the prediction of CL; and
iv. Geographic Information System (GIS) techniques will be used in presentation of CL of the various regions in a map format.

1.5 Facilities used

The facilities to be used are:

i. UMaT Library for academic resources and reference materials;
ii. Online databases for accessing research articles;
iii. SmartPLS 4, IBM SPSS and Jupyter Notebook for modelling and prediction of CL;
iv. ArcGIS 10.7 for preparation of Maps; and
v. A Personal Computer for data analysis, interpretation and report writing.

1.6 Organization of the report

There are six chapters in total of this project. The problem statement, projects objectives, project methodologies as well as facilities and expected outcomes, are all presented in Chapter 1. The chapter 2 contains relevant literature about the study area. Important previously work on related areas regarding the study was examined and detailed in chapter 3. The methodologies and techniques to achieve the project objective are also presented in chapter 4. Chapter 5 contains the result of the methodologies employed as well as the discussions. Finally, the chapter 6 discusses the project including the required suggestion.

CHAPTER 2
THE STUDY AREA

2.1 Introduction

The chapter provides information of the study area. Its gives the geographical boundary, the topography, the climate, and demographics in which the project and data collected is confined.

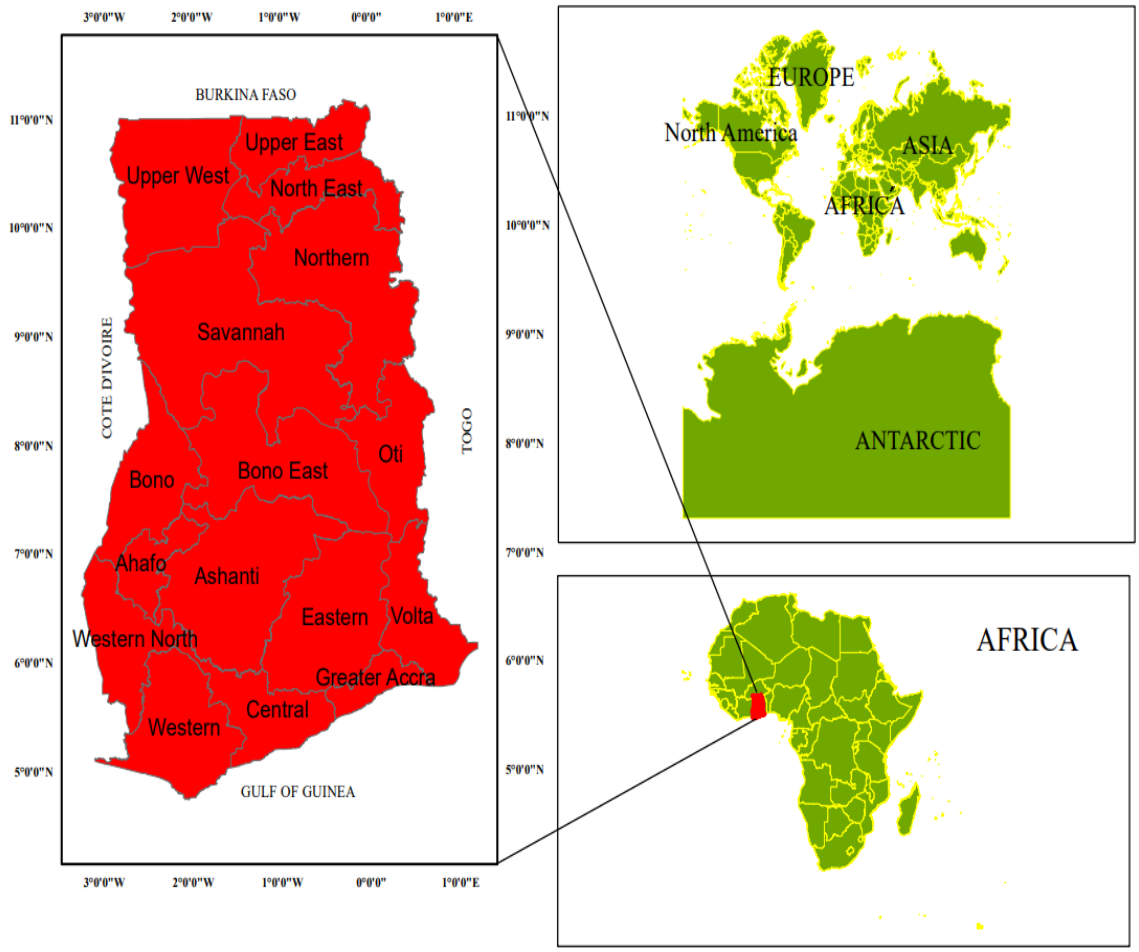
2.2 Location and Size

This study covers the entire region of Ghana see (figure 1.1). Ghana is a nation that lies between the Atlantic Ocean and the Gulf of Guinea, in sub-region of West Africa (Opoku, 2018). Covering a total land mass of 238,535 $\boldsymbol{km^2}$ (92,099 sq. mi), Ghana shares boundary with Burkina Faso, Togo, Ivory Coast, and the Gulf of Guinea and the Atlantic Ocean in the south. The country's terrain is diverse, with tropical rainforest, savanna and coastal plains. Ghana has approximately 32 million people with over 100 ethnic groups, but the official language is English. The culture in Ghana is diverse, with a strong emphasis on traditional music, dance and art (Opoku, 2018). Ghana has a rich and complex history with information of habitation of human around the Stone Age (Boafo, 2018).

2.3 Topography

Ghana's topography can be broadly classified into four main regions including Coastal Plains, the coastal plains of Ghana stretch along the Gulf of Guinea and are characterized by a narrow strip of land, typically less than 50 km wide (Dickson *et al*, 2004). Forest Zone, the forest zone covers the southwestern part of Ghana and is characterized by dense tropical rainforests, Mountain Ranges including Kwahu Plateau, Togo-Atakora Mountains, Rivers and Lakes, including Volta River (Hall *et al*, 1981).

Fig 1 Map of Study Area (Anon, 2022)



2.4 Climate

Ghana boasts a tropical environment marked by high temperatures and high humidity all year long since it lies in the tropics. The country's location close to the equator and its proximity to the Atlantic Ocean shape its temperature (Acheampong, 2017).

Ghana's annual temperature ranges from March and April as the greatest temperatures while August and September have the lowest temperatures. Over the year, Ghana's average temperature runs from 22 °C to 32 °C (72 °F to 90 °F) (Dickson *et al*, 2004).

Ghana has an average relative humidity ranging between 60% and 80%. The country experiences a high level of humidity along the year. The southern part records the highest humidity mostly in the coastal areas (Acheampong, 2017).

Ghana experiences two distinct rainy seasons, from April to July, has major falls so mostly termed as the main rainy season, September to November also termed the minor rainy season. The rainfall pattern varies across the country, with the highest rainfall recorded in the southwestern part of the country (Dickson *et al*, 2004).

An average of average of 6 to 8 hours of sunshine is recorded most of the days in Ghana. The northern part of the country specifically the Savanna areas, enjoys the most hours of sunshine (Acheampong, 2017).

2.5 Sample Size

This study collected data using a structured questionnaire. The poll asked respondents on factor including customer satisfaction, service quality, customer orientation and use of social media. Each major factor allows respondent to input value from a scale of 1-10 there, based on their preference. The questionnaire was open to all clients, customers and stakeholders of the CSAU. A total of 300 samples were used in the study process with respondent from all across the country.

CHAPTER 3
LITERATURE REVIEW

3.1 Introduction

This chapter provides a comprehensive literature review of the major methods and model that were used in this study.

The Land Administration System (LAS), a branch of the lands commission of Ghana deals with transaction mainly on lands. LAS provide various services to customers through a service called Clients Satisfaction Access Unit (CSAU). The CSAU was designed to aid in transparency in land transaction and enhance the efficiency of the service. Customer loyalty a major factor to the success of the continual use of the service is at stake, since no previous work has been done to assess and predict for such a purpose regarding the CSAU. The review was done purposely to ascertain relevant methods and approaches in counterattacking such an issue. The main review was therefore done on Structural equation modelling (SEM), Recurrent Neural Network (RNN) and Support vector machine (SVM) as well as methods to evaluate the models.

3.2 Customer Loyalty

The willingness and consistency of customers to rely firmly on a particular firm while referring it to others is termed Customer Loyalty. Huang *et al*, (2019) defined Customer loyalty can be termed as preventing promotions and offers from competiting businesses, making purchase of a certain goods and services, as well as offering referrals to others. The two components for measuring consumer loyalty are behavioral and attitudinal loyalty (Samuel *et al*, 2024). Customer loyalty refers to the likelihood of customers to use a service for a period of time. It's a measure of how well a company has earned the trust and satisfaction of its customers.

Customer Service and related factors can be measured once the factor that affects customer loyalty has been detected. Many quantitative methods are put in place today to quantify and predict customer loyalty respectively. Depending on accurate factors, the service or product owner is able to perform a prediction and learn from it which factors to be improved (Chou *et al*, 2014). The study comprises of one dependent variable, Customer Loyalty and four major independent variables namely, Service Quality, Customer

satisfaction, Customer Orientation and social media. Each of the major independent variable has two or three sub variables. The hypothesis derived for these projects include,

H1: Customer orientation has positive effects on customer satisfaction

The extent to which customers becomes satisfied with a service can be determined with customer satisfaction. Customers who are happy with a service are more likely to recommend it to others, hence introducing more customers. Customer orientation also has an effect on customer satisfaction, according to Li *et al*, (2022), who evaluated 150 studies to look at the relationship existing between orientation and relative activities.

H2: Service quality has a positive impact on customer satisfaction

Service quality characteristics have direct effects on customer satisfaction. Many research labels that customer satisfaction is affected by Service quality. Yap (2015) indicated that, customer ability to purchase goods and services is strongly dependent by their satisfaction level as well as the quality of the service they receive. The results of other studies also support that of Syahputra *et al,* (2022) and that of Adriani *et al*, (2023) who have established that Service quality is positively affected by customer satisfaction.

H3: Customer Satisfaction has a positive effect on Customer Loyalty

Customer satisfaction has a strong effect on customer loyalty, according to the study by Khairunnisa *et al*, (2020). The findings of other studies and research are also in line with Lina (2022), who conclude that Customer Satisfaction has positive impacts on Customer Loyalty Lina (2020).

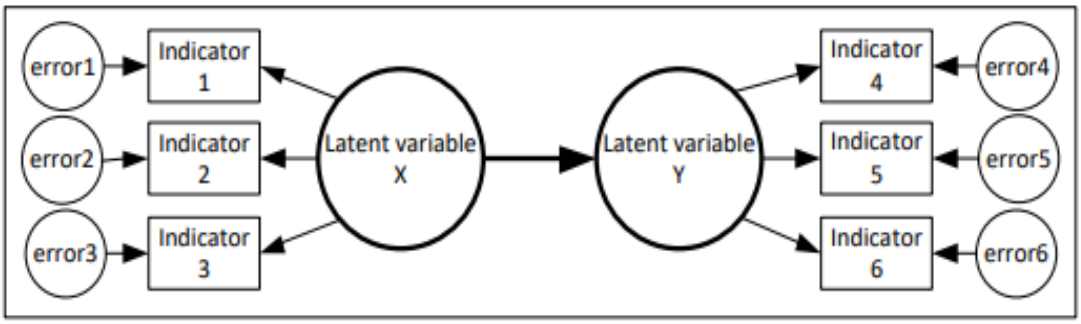
H4: Social media has a positive effect on customer satisfaction

To be able to assess the impact of social media on customer satisfaction, Kim *et al,* (2022) reviewed 25 studies to assess the relationship that exist between customer satisfaction and Social media. The study's finding concluded that Social media can positively affect customer satisfaction (Kim *et al*, 2022).

3.3 Structural Equation Model

SEM has been one of the most popular tools for determining and measuring the correlations between variables since it was first proposal in 1921. SEM enables a detailed analysis of the relationship between variables by combining various components of factor analysis, regression analysis and path analysis (Hadiyat, 2019).

Fig 3 Simple SEM Model (Hadiyat, 2019)



SEM has been applied in many industries to predict loyalty and satisfaction based on the linear relationships that exist between these variables. These industries include the hotels, telecommunication banking sector, industry, and supermarket

In 2019, M. A Hadiyat published a paper titled: "Combined structural equation modelling – artificial neural networks model for predicting customer loyalty". 130 customers of a hotel filled a questionnaire which aid in the prediction of the loyalty of the customers using SEM and ANN techniques. This research concluded that hybrid SEM-ANN is successful in predicting the customer loyalty thereby making rooms for improvement strategies (Hadiyat, 2019).

In the telecommunication industry several works has been done including the one published by Oduro *et al*, (2018) titled "determinants of customer satisfaction in the telecommunication industry in Ghana". This research considered the factors including customer expectation, relationship quality, perceived value, perceived quality, and customer loyalty. The SEM and the American Customer Satisfaction Index (ACSI) model were used. This study concludes that when managing customer connections, MTN Ghana should consider how customers understand the quality of some product and services

offered as well as the extent of impact of service quality and customer loyalty (Oduro *et al*, 2018).

Several works has also being done regarding SEM in the business research fraternity including the work publishes by Jun Cui title "Comparative Analysis of PLS-SEM, SEM-ANN, and fsQCA in Modeling Complex Causal Relationships in Social Sciences and Business Research". This study findings indicated that, fsQCA allows for a new approach to determine causal configurations in small sample studies, SEM-ANN improved in incorporating non-linearities and adaptability whiles PLS-SEM improved at modelling the relationship that exist variables in larger datasets (Jun, 2024)**.**

SEM as used in various fields was also used in the hospitality and tourism industry. Among recently published papers from Vietnam is a paper titled "SEM-ANN analysis to examine the green consumption behavior of tourists in Vietnam". This research combined a hybrid SEM-ANN and the Value-Attitude-Behavior (VAB) to investigate the green consumption behavior of tourists. This research concluded by giving credit to the high predictive power of the combined method demonstrating the reasonableness of the SEM-ANN hybrid approach (Nguyen *et al*., 2025).

In the field of digital communication similar work using SEM was published titled "Exploring pathways to comprehension performance in Multilanguage smart voice systems: insights from Lasso regression, SEM, PLS-SEM, CNN**,** and BiLSTM". This research used SEM to assess the primary factors impacting the performance of comprehension in Multilanguage smart voice systems (Entong *et al,* 2024).

This project hence aims to use SEM in assessing the primary and key determinant of customer loyalty of Land administration system in Ghana.

With regards to confirmatory studies, the fitness between a proposed model and the dataset is the main objective. The covariance- based SEM (C-SEM) is used mostly. C-SEM is very used frequently for hypothesis testing because of its ability to reduce the difference existing between observed and computed covariance matrices (Byrne *et al,* 2016).

3.4 Recurrent Neural Network (RNN)

Recurrent Neural Networks (RNNs), was designed for handling sequential data. RNN comprises of key component including, Recurrent Neural Networks (RNNs) are specialized models for sequence modelling (Anon, 2025). In brief, RNNs process sequential data by iteratively computing hidden states as follows:

$$h_t = f_\theta(h_{t-1}, x_t). \qquad (1)$$

Where t denotes the step index, x represents a token; h represents the hidden state, and f_θ is a neural network parameterized by θ. The value of the initial hidden state $\boldsymbol{h_o}$ is typically learned via back propagation.

The exploding and vanishing gradient problem are among the major problems that affects the RNN in literature. Exploding and Vanishing Gradient Problem (EVGP), makes RNNs challenging to train because their loss gradient back propagation tends to saturate or diverge during training (Schmidt *et al,* 2019).

To address the limitations of standard RNNs, particularly the vanishing gradient problem Long-Short-Term-Memory (LSTMs) was introduced (Hochreiter *et al,* 1997). To be able to selectively maintain, update and forget information over long sequences, LSTMs incorporate special gating techniques. Vikram et al, (2024) used the LSTMs to predict yield of crop which produced the best model for the prediction

A computational method that establishes a recurrent neural network's weights and biases w ithout the inclusion of gradient-based methods was also introduced to mitigate the Exploding and Vanishing Gradient Problem. It was found that using such method improved the training time and predicting accuracy as compared to gradients-based methods (Erik *et al*, 2023).

LSTM has been utilized in many researches to predict customer satisfaction and loyalty. LSTM was compared and utilized in forecasting retail demand. The research concluded that LSTM outperformed the other models that are best in capturing the complex and temporal dependencies of the study with an R^2 of 0.90, RMSE of 14.67 and MAE of 9.53, and excelling in capturing temporal dependencies and complex demand patterns (Shujan *et*

al, 2024). This project therefore seeks to utilize LSTM for prediction of customer loyalty of the Client Satisfaction Access Unit.

3.5 Support Vector Machine (SVM)

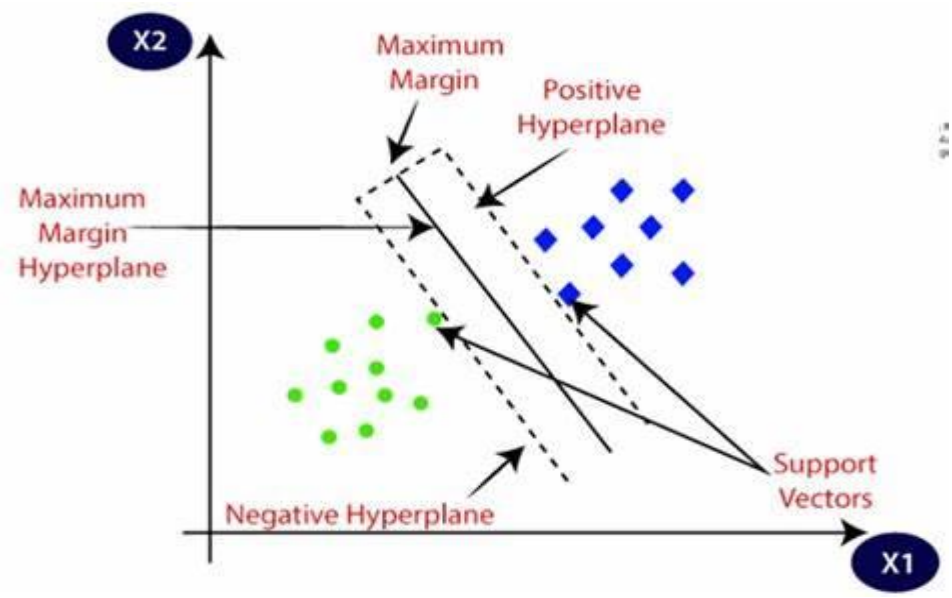
A Supervised machine learning method suited for classification and regression applications. SVM is well-suited for classification task while at same time takes care of regression problems. The best hyper plane in an N-dimensional space to provide data points to different classes is the best goal of an SVM (Anon., 2020). Support vector machine is used to execute non-linear classification effectively through the kernel trick, which transforms their domain into high dimensional feature spaces (Ghadar *et al*, 2017). Specifically, a support vector machine defines a hyper plane in an infinite-dimensional space that even higher than the dimension of samples to facilitate multi-task classification, regression, and other tasks (Ghadar *et al*, 2017).

SVM has been a powerful tool used for robust prediction in many fields. SVM has been employed to classifier sleep disorder with an estimated accuracy of 92% indicating that Support Vector Machine can enhances the diagnosis of sleep disorders, allowing for early intervention and better care for patients (Nenden *et al,* 2025).

(Syakira *et al,* 2023) did a study to student graduation using the SVM and the study concluded that the support vector machine model's accuracy level is 71.20%, f1-score of 82% and recall of 72%.

A study done in the Airline Hospitality services for the Prediction of Customer Satisfaction concluded that SVM outperformed with the best performance of prediction. The performance average accuracy, precision and F1-score are 93, 93 and 92% respectively (Adedayo *et al,* 2024).

Fig 4 SVM Architecture (Anon., 2022)



There are some kernel functions built in the program but it is also able to choose the preference that fits and data set for compute a precise result.

3.6 Model Assessment

Data form the questionnaire was used in the modelling of the SEM, using Amos the fitness of the hypothetical model was used in ascertaining the fitness of the model. The confirmatory factor analysis was as well applied, confirmatory factor analysis (CFA) computes for the calculations of both the measurement and structural models simultaneous, enabling accurate estimation of the latent variable's relationships (Jöreskog *et al,* 2018). The software's capacity to manipulate large datasets and non-normal data distributions can contribute to the best option for interpreting the interrelationships existing between the variables. The Chi-Square statistic, Root Mean Square Error of Approximation (RMSEA), Comparative Fit Index (CFI), and Tucker-Lewis Index (TLI) are the indices used in assessing the model fits of the dataset. This index allows for a critical evaluation of the model. A higher fit is indicated by a lower mark in the Chi-Square Statistics, as it's used in evaluating the overall fitness of the model. The RMSE, which evaluates the model's approximation errors of the degrees of freedom is well fitted when a value less than 0.06 is achieved. CFI and TLI compare the fitness of the model to a null model and a value near 0.95 is considered best model fit (Unegbu *et al.*2025).

The model assessment for Support vector machine (SVM) and Recurrent Neural Network (RNN) was mainly through absolute and squared error matrices including, MAE, MAPEE, MSE among others.

CHAPTER 4
MATERIAL AND METHODS USED

4.1 Materials

4.1.1 Data

The dataset used for this project were obtained through the drafting and dissemination of questionnaire. This questionnaire data contains the entire relevant determinant needed for predicting the customer loyalty of the LAS.

4.1.2 Software Used

Table 4.1.2 Softwares Used

Software Used	Application
Microsoft Excel 2019	For data cleaning
IBM SPSS AMOS	For SEM Modelling
Orange Data Mine	For SVM modelling
Google Colab	For ANN Modelling

4.2 Methods

4.2.1 SEM Modelling

The study takes on predetermined variables in modelling the Customer Loyalty of the LAS in Ghana. One dependent and four independent variables were measured through a questionnaire filled by 300 individual who have been using the system. SEM uses a linear statistical approach to identify the complex and intrinsic relationship that exist between this variable. Based on these statistical methods the fitness of the modelled is tested though the Akaike's information criterion (AIC) as well as the confirmatory factor analysis (Hadiyat, 2020). Significant variables chosen based on the model fit were used as the input for the modelling of the RNN and SVM.

4.2.2 Recurrent Neural Network Modelling

The relevant variables from the SEM modelling were used as input for the RNN modelling. Long Short-Term Memory (LSTM) was used in other to solve the vanishing gradient problem. Important libraries Such as pandas for data manipulation, numpy for numerical computations, and tensor flow for building the LSTM model was imported. The data was then loaded. Preprocessing was done through scaling/normalizing the features, encoding categorical variables, and managing missing values. The data was splitted into training and testing sets (70, 30). The data was reshaped to have a shape of samples, timesteps and features. The LSTM model was then built using the following code. The model was compiled using a suitable optimizer and loss function. The model was then trained and evaluated.

4.2.3 Support Vector Machine Modelling

The relevant inputs from the SEM modelling were used as input in modelling the SVM. Data Preparation including cleaning missing values and scaling or normalizing the features were performed on the dataset. The significant input from SEM was then split into training (80%) and testing sets (20%). The kernel selected for the project was linear excluding polynomial, or radial basis function (RBF). However, these parameters were later tuned to aid in the accuracy of the model.

The dataset was loaded into Orange by clicking on the "File" menu, select "Open", and choose my dataset file. After loading the data, preprocessing was done to through scaling/normalizing the features, encoding categorical variables, and managing missing values using "Data" widget. The preprocessed data was partitioned into sets for testing and training using the "Data" widget in the order 80% of the data for training and the remaining 20% for testing. An SVM model was created using the by dragging and dropping the widget onto the workflow canvas. Configuration of the SVM model was done by selecting the kernel type, regularization parameter (C), and other relevant hyper parameters. Train the SVM model using the training data by connecting the to the widget. The performance of the model was assesses using the testing data. "SVMLearner" widget was connected to the "Test & Score" widget and click on the "Evaluate" button.

Finally, the result was visualized using various visualization widgets, such as the "Confusion Matrix" widget or the "ROC Curve" widget. The Model was evaluated using

metrics like ROC-AUC, F1-score, recall, accuracy, and precision. To get the highest accurate model and the performance of the SVM model the hyper parameters were adjusted. The Selected best-performing model based on evaluation metrics was then saved.

4.3 Model Assessment

4.3.1 Confirmatory Factor Analysis

Verifying if the data fit a proposed measurement model is the aim of confirmatory factor analysis. Theory and/or prior analytical study serve as the foundation for this proposed model (Anon., 2022).

In order to arrive at the relationships between sets of latent and observable variables, factor analysis is used. Exploratory factor analysis is used when there are unclear or unknown relationships between the latent and observable variables. To determine how and to what degree the observable variables are connected to their underlying factors, factor analysis is performed (Rajendra, 2024).

The study employed customer loyalty as the dependent variables and customer orientation, customer satisfaction, service quality and social media as the major independent variables affecting the customer loyalty of the LAS. The hypotheses developed were then tested for its fitness. The fitness explains how effectively the covariance between each item and measure in the model was captured by the suggested model. The model will be rejected if the statistical assessments of model fit show that it does not fit well. Some items measure numerous factors, which can then be manipulated to co-vary each other while still checking for fitness, which could be the cause of a poor fit.

Following a model fit test, a number of values and indicators were employed in AMOS to evaluate the model's goodness of fit. Here are a few of the most important metrics and values: Chi-Square ($\chi 2$): The gap between the observed and expected covariance matrices is measured by this statistic. A better match is indicated by a smaller $\chi 2$ value. Degrees of Freedom (df): This is the number of parameters estimated in the model: This is the probability of observing a $\chi 2$ value as extreme or more extreme than the one observed.

A p-value more than 0.05 shows that the model fits the data. Root Mean Square Error of Approximation (RMSEA): measures the average difference between the observed and expected covariance matrices. An RMSEA value of 0.06 or less represents a good fit.

The model's fit in relation to a baseline model is gauged by the CFI. A good match is indicated by a value of 0.95 or higher. Similar to the CFI, the Tucker-Lewis Index (TLI) accounts for the number of parameters estimated in the model. An excellent match is indicated by a TLI value of 0.95 or higher.

An indicator of the mean variance between the actual and expected covariance matrices is called the Standardized Root Mean Square Residual, or SRMR. A good match is indicated by an SRMR value of 0.08 or below. These numbers and metrics offer a thorough assessment of the model's fit and aid in finding out whether the model accurately depicts the data. Good fit is indicated by $X2/df < 3$, $CFI > 0.95$, $TLI > 0.95$, $RMSEA < 0.06$, and $SRMR < 0.08$ (Anon, 2020).

4.3.2 Akaike's Information Criterion (AIC)

The Akaike's Information Criterion (AIC) is a statistical approach for evaluating the quality of a model for a given set of data. It's a popular criterion for model selection, especially in situations where you have multiple models and want to choose the best one.

AIC measures the relative loss of information when a given model is used to approximate the true underlying process that generated the data, it estimates the amount of information lost when a model is used to simplify the complexity of the data.

AIC is calculated using the following formula:

$$AIC = 2k - 2\ln(L) \qquad (1)$$

Where:

K is the total number of parameters;

L is the maximum likelihood of the model.

A smaller AIC value shows that the model is a better fit to the data. In general, a difference of 2-3 units in AIC values between two models is considered significant. The model with the least AIC value is generally considered the optimal model. AIC is

relatively easy to calculate, especially when compared to other standards for choosing the models like the Bayesian Information Criterion (BIC). AIC values are easy to interpret, with lower values indicating better model fit. AIC assumes that the model parameters are estimated using maximum likelihood estimation.

4.3.3 RNN model Analysis

The RNN Model was assessed for accuracy of the model on the test dataset. Evaluation models used includes, evaluating the loss function such as mean squared error on the test dataset as well as the precision of the model,

The model performance was also assessed through training and Validation Loss Plots: where the training and testing dataset was plotted for loss against epochs to check for over fitting or under fitting.

Cross-Validation was Perform to computes for the model's performance on the abstract data as well as hyper parameter tuning to optimize the model's performance.

4.3.4 SVM model Analysis

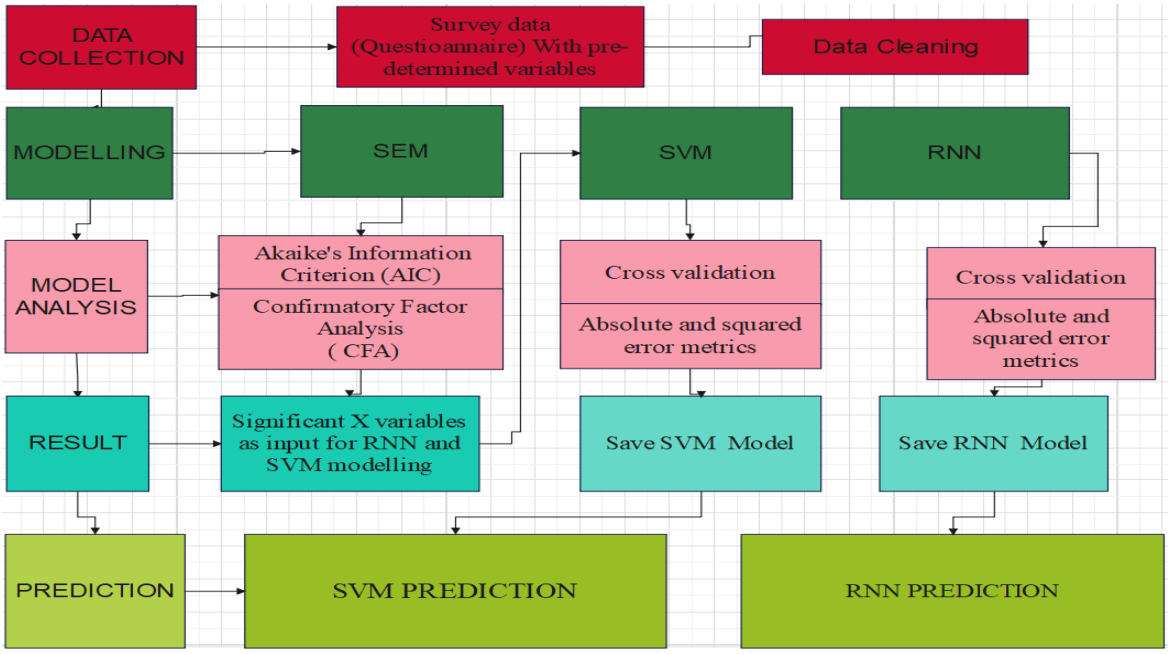
The performance of the SVM model was assessed using the "Test & Score" widget. The classification results of the SVM model was assessed using the "Confusion Matrix" widget. By connecting the "Test & Score" widget to the "Confusion Matrix" widget and click on the "Show" button. The feature importance of the SVM model was assessed using the "Feature Importance" widget, by connecting the widget to the "Feature Importance" widget and click on the "Show" button.

4.4 Proposed Framework

The framework for the methods used is grouped into five major categories namely data collection, data modelling, evaluation of models and analysis, results generated and prediction. The data collection involves the designing of questionnaire containing already determined variable. The questionnaire was then distributed to receive inputs from customers of the CSAU. The data obtained was cleaned and prepared to be used for the modelling using SEM. Modelling stage of the study employs the data obtained from the questionnaire to first assess the fitness of the independent variable (X) and dependent

variable (Y) through Confirmatory Factor Analysis (CFA). The significant X variables was now used as an input in the modeling of the SVM and RNN. The overall data was partitioned into training and testing data in the ratio 80: 20. Data normalization was used to scale the input into a scalable interval (1, 0). Evaluation and analysis stage of the sturdy seeks to evaluate the accuracy of the SEM using the Akaike's International Criterion (AIC). The SVM and RNN models will be evaluated using the absolutes and squared errors metrics. The Cross validation methods were also used as evaluation methods, where the data was randomly partitioned without any defined order or standardized methods. The results stage gives an account of the output obtained from the modelling of SEM, SVM and RNN models. The significant X variables was saved as the results of the SEM modelling whiles the best tested model was saved for both SVM and RNN. The final stage is the prediction. Here new customers, fills questionnaire containing all the relevant X variables. This Input from new customers was fed to the saved model. The model then predicts whether that customer is loyal or not depending on the input.

Fig 5 Proposed Framework Used

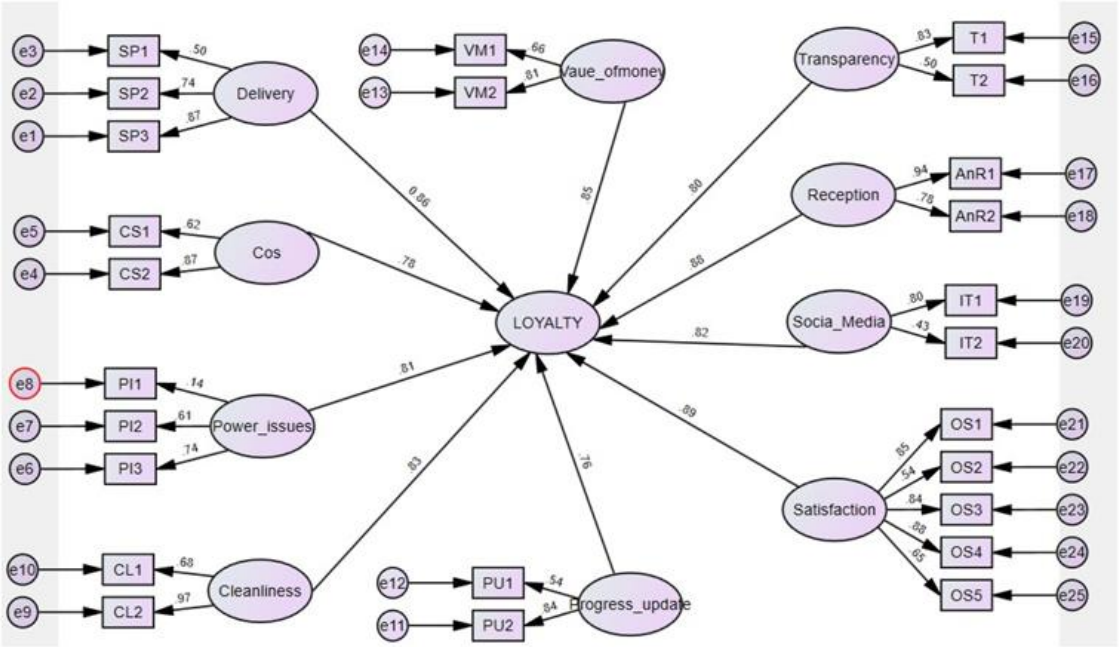


CHAPTER 5
RESULTS AND DISCUSSION

5.1 SEM Modelling

The structural equation model aims to analyze the relationships that exist between observed and unobserved variables. The variable selected as significant based on the factor loading as shown in table 5.1 below. The questionnaire was measured in a likert scale of 10, with values equal or closer to ten regarded as significant. The SEM path model is illustrated in figure 5.1 below.

Figure 5.1 SEM Path Model



5.2 SEM Analysis

The relevant variable selected was based on statistical measures such as TLI which measures how well structure equation fits the observed variables, CFI, RMSEA, SRMR and the variables respective Factor weighting. Table 5.2 gives the factor weighting exhibited by each variable; The TLI index recorded is 0.976, CFI value of 0.983 which assesses how well the model fits the observed dataset and RMSEA of 0.022 which measures the discrepancies between the observed covariance matrix and PCLOSE value of 0.978 which refers to the p-value for rejecting the null hypothesis.

Table 5.1 Variable Predictor Response

ID	Variable	Code	Significance	Weights
1	Orientation and Reception	OnR	YES	0.88
2	Social Media Platform	SM	YES	0.82
3	Transparency	T	YES	0.80
4	Progress Update	PU	Y ES	0.76
5	Service delivery	SP	YES	0.86
6	Cost of Service	CS	YES	0.78
7	Customer Satisfaction	OS	YES	0.89
8	Environment Cleanliness	CL	YES	0.83
9	Power Issues	PI	YES	0.81
1O	Staff Value of Money	VM	YES	0.85
11	Loyalty (Target)	LOYALTY		

5.3 SVM Modelling and Prediction

Significant predictors from the SEM modelling were used as the input for the SVM modelling (See Appendix B). Ten variables were used for the prediction whiles targeting loyalty. The prediction widget was used to predict the target variable (loyalty) whiles using the test and score was used to access the accuracy of the model, see (figure 5.3) below. The predicted SVM with respects to actual and errors is illustrated in (fig 5.2).

Fig 5.2 SVM prediction

	SVM	error	LOYALTY	ONR	C	SM	T	PU	SQ	CS	Csat	CL	PI
1	6.281	0.298	5.983	0.250	0.250	0.0	0.750	0.5	0.5	0.5	0.250	0.5	0.5
2	6.921	0.300	6.621	0.5	0.5	0.5	0.5	0.5	0.250	0.5	0.5	0.5	0.0
3	7.448	0.174	7.274	0.5	0.5	0.5	0.750	0.5	0.250	0.5	0.5	0.5	0.5
4	7.319	0.303	7.016	0.5	0.5	0.5	0.750	0.750	0.250	0.750	0.5	0.5	0.5
5	7.057	0.299	6.758	0.5	1.0	0.0	0.5	0.5	0.0	0.5	1.0	0.5	0.5
6	7.265	0.301	6.964	0.0	0.750	0.750	0.0	0.750	0.0	0.750	0.750	0.5	0.5
7	6.903	0.301	6.602	0.750	0.5	0.5	0.750	0.0	0.5	0.5	0.5	0.5	0.750
8	7.244	-0.301	7.545	0.5	0.250	0.0	0.5	0.5	0.250	0.5	0.250	0.5	0.750
9	8.220	-0.302	8.522	0.5	0.5	0.5	1.0	0.750	0.0	0.5	0.5	0.5	0.5
10	7.335	-0.111	7.446	0.250	0.5	0.5	0.750	0.5	0.5	0.0	0.5	0.5	1.0
11	7.044	0.303	6.741	0.750	0.5	0.250	0.5	0.5	0.250	0.250	0.5	0.5	0.250
12	7.033	0.301	6.732	0.250	1.0	0.750	0.5	0.750	0.0	0.5	1.0	0.5	0.5
13	7.370	0.028	7.342	0.5	0.5	0.250	0.5	0.5	0.250	0.5	0.5	0.5	0.5
14	7.407	-0.299	7.706	0.5	0.250	0.5	1.0	0.750	0.250	0.250	0.250	0.5	0.750
15	7.328	0.045	7.283	0.250	0.5	0.5	0.5	0.5	0.0	0.5	0.5	0.5	0.250
16	7.362	0.087	7.275	0.5	0.750	0.750	1.0	0.250	0.250	0.250	0.750	0.5	0.5
17	7.906	-0.299	8.205	0.250	0.750	0.0	0.750	0.750	0.250	0.250	0.750	0.5	0.5
18	6.683	0.301	6.382	0.250	0.750	0.750	0.5	0.5	0.0	0.5	0.750	0.5	0.5
19	6.719	0.297	6.422	0.5	0.250	0.250	0.5	0.5	0.250	0.5	0.250	0.5	0.5
20	7.296	0.022	7.274	0.5	0.5	0.5	0.250	0.250	0.5	0.5	0.5	0.5	0.250

The regression metrics MAPE, $\mathbf{R^2}$, MAE, RMSE and MSE with values of 0.030, 0.842, 0.218, 0.242 and 0.058 respectively. The MAPE calculates the percentage-based error, $\mathbf{R^2}$ indicating how well the model explains the variance in the data. MSE measures average squared difference between predicted and actual values

Fig 5. 3 Accuracy Assessment matrices

Model	MSE	RMSE	MAE	MAPE	R2
SVM	0.058	0.242	0.218	0.030	0.842

Fig 5.4 Feature statistics

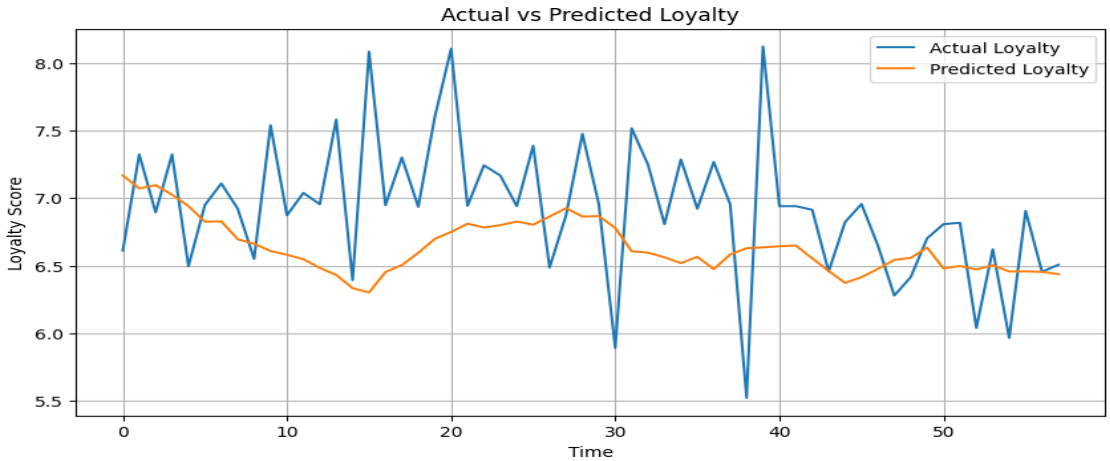
	Name	Distribution	Mean	Mode	Median	Dispersion	Min.	Max.	Missing
N	ONR		0.363	0.5	0.250	0.751	0.0	1.0	0 (0 %)
N	C		0.410	0.5	0.5	0.589	0.0	1.0	0 (0 %)
N	SM		0.356	0.5	0.250	0.730	0.0	1.0	0 (0 %)
N	T		0.316	0.333	0.333	0.810	0.0	1.0	0 (0 %)
N	PU		0.5	0.5	0.5	0.473	0.0	1.0	0 (0 %)
N	SQ		0.431	0.5	0.5	0.544	0.0	1.0	0 (0 %)
N	CS		0.457	0.5	0.5	0.471	0.0	1.0	0 (0 %)
N	Csat		0.410	0.5	0.5	0.589	0.0	1.0	0 (0 %)
N	CL		0.489	0.5	0.5	0.402	0.0	1.0	0 (0 %)
N	PI		0.421	0.5	0.5	0.640	0.0	1.0	0 (0 %)
N	LOYALTY		6.87404	7.274	6.83250	0.09814	5.538	9.011	0 (0 %)

The feature statistics helps assess the quality and relevance of the features used in the modelling as well as checking outliers such as such as min/max values of the datasets used.

5.4 ANN Modelling and Prediction

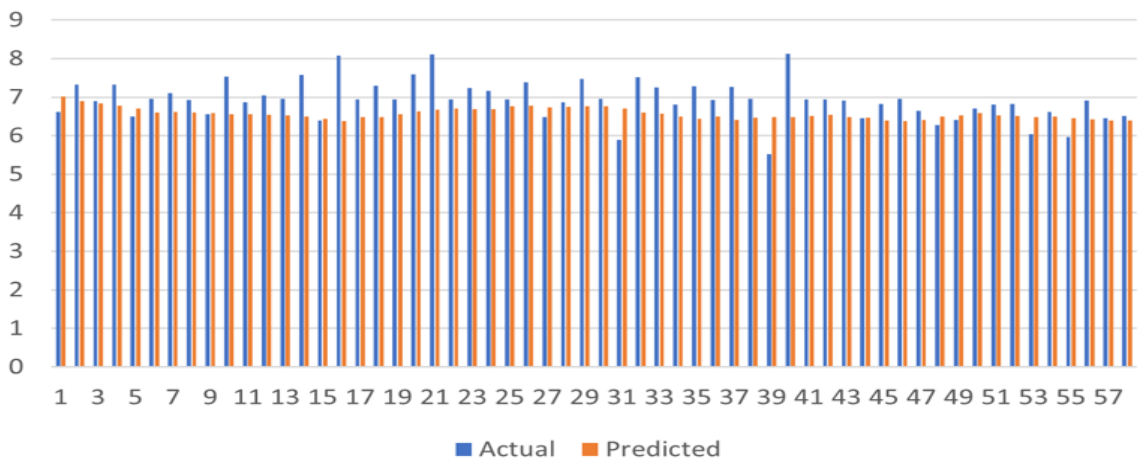
The performance of the Long Short-Term Memory (LSTM) model was evaluated using statistical indicators such as MAE, $\mathbf{R^2}$ and RMSE. The values are 0.4904, 0.6145 and 0.5105 respectively. The predicted loyalty against actual loyalty with respect to time is shown in (figure 5.5) below

Fig 5.5 Loyalty against time



The actual loyalty to predicted loyalty shows a good fits and performance accuracy. The average difference between the predicted and actual loyalty (0.234) lies within the allowable range of tolerance of deviation. See fig 5.6 below

Fig 5.6 Actual to predicted loyalty



5.5 Discussion

The Structural model proved ten out of twelve variables as relevant for the prediction of customer loyalty. Loyalty scores obtained was not of the best scores since the required number of samples was limited. Both the SVM and LSM model used in the modelling and prediction of customer loyalty were tested using statistical indicators. The SVM model with R2, MAPE, MAE, RMSE and MSE values of 0.842, 0.030, 0.218, 0.242 and 0.058 respectively whiles the LSTM model statistical indicators such as MAE, RMSE and R2, had values of 0.4904, 0.6145 and 0.5105 respectively. In the feature statistics table, min/max illustrate the central tendencies and range of numerical feature. Mode indicating the most frequent value with value of 7.274 for loyalty (target) revealing the dominant or most common loyalty level.

CHAPTER 6
CONCLUSION

The project was aimed to predict customer loyalty based on twelve observable variables. The variables obtained for the prediction of customer loyalty was in a total of ten after structural equation modelling. The relevant variable was then used as input for the SVM and LSTM modelling. The ten various as input whiles the variables scores obtained from structural model as target (loyalty). The SVM showed performance accuracy of 79% with R2 of 0.944. The predicted results showed insignificant errors. The SVM ability to handle high dimensional and non-linear data was showcased. The LSTM on the other hand performed but not as accurate as the SVM. The R2 value is 0.5105 which shows moderately accuracy. The LSTM model predicted customer loyalty for ten samples and had an RMSE of 0.6145. Both model predicted per the data obtained, however the medium performance of the LSTM is believed to be as a result of the low scores obtained from the SEM modelling. In literature fitness in SEM model requires higher samples.

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Appendix A: Factor Analysis Using Amos

Model Fit Summary

CMIN

Model	NPAR	CMIN	DF	P	CMIN/DF
Default model	32	52.338	46	.241	1.138
Saturated model	78	.000	0		
Independence model	12	447.144	66	.000	6.775

RMR, GFI

Model	RMR	GFI	AGFI	PGFI
Default model	.047	.971	.951	.573
Saturated model	.000	1.000		
Independence model	.139	.777	.736	.657

Baseline Comparisons

Model	NFI Delta1	RFI rho1	IFI Delta2	TLI rho2	CFI
Default model	.883	.832	.984	.976	.983
Saturated model	1.000		1.000		1.000
Independence model	.000	.000	.000	.000	.000

Parsimony-Adjusted Measures

Model	PRATIO	PNFI	PCFI
Default model	.697	.615	.685
Saturated model	.000	.000	.000
Independence model	1.000	.000	.000

NCP

Model	NCP	LO 90	HI 90
Default model	6.338	.000	28.380
Saturated model	.000	.000	.000
Independence model	381.144	317.899	451.879

FMIN

Model	FMIN	F0	LO 90	HI 90
Default model	.176	.021	.000	.096
Saturated model	.000	.000	.000	.000

Independence model	1.506	1.283	1.070	1.521

RMSEA

Model	RMSEA	LO 90	HI 90	PCLOSE
Default model	.022	.000	.046	.978
Independence model	.139	.127	.152	.000

AIC

Model	AIC	BCC	BIC	CAIC
Default model	116.338	119.268	234.645	266.645
Saturated model	156.000	163.141	444.373	522.373
Independence model	471.144	472.242	515.509	527.509

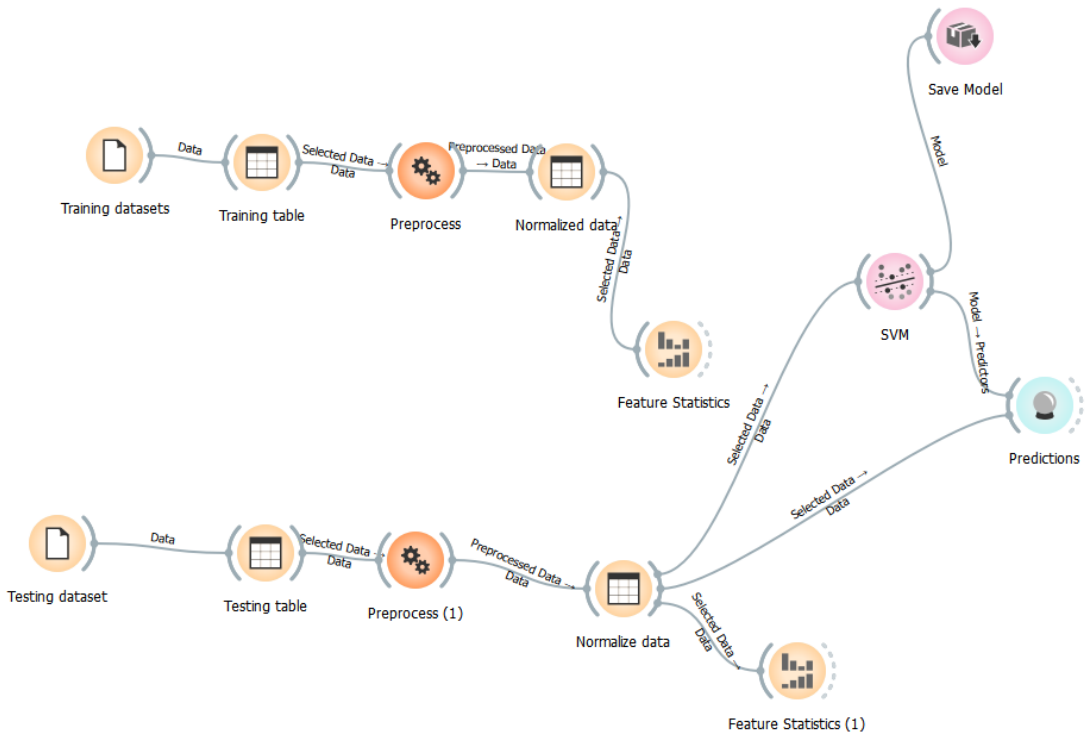
ECVI

Model	ECVI	LO 90	HI 90	MECVI
Default model	.392	.370	.466	.402
Saturated model	.525	.525	.525	.549
Independence model	1.586	1.373	1.825	1.590

HOELTER

Model	HOELTER .05	HOELTER .01
Default model	357	405
Independence model	58	64

Appendix B: Support Vector Machine Workflow in Orange



Appendix C: Long Short-Term Memory (LSTM) Implementation in Google Colab

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error,
r2_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
import matplotlib.pyplot as plt

# 1. Load your dataset
data = pd.read_csv('LOYAL.csv')

# 2. Select your 10 SEM-identified features
selected_features = ['ONR', 'C', 'SM', 'T', 'PU',
                     'SQ', 'CS', 'Csat', 'CL', 'PI']

X = data[selected_features].values
```

```
y = data['LOYAL'].values

# 3. Scale the data
scaler_X = MinMaxScaler()
scaler_y = MinMaxScaler()

X_scaled = scaler_X.fit_transform(X)
y_scaled = scaler_y.fit_transform(y.reshape(-1, 1))

# 4. Create sequences for LSTM
def create_sequences(X, y, time_steps=10):
    Xs, ys = [], []
    for i in range(time_steps, len(X)):
        Xs.append(X[i-time_steps:i])
        ys.append(y[i])
    return np.array(Xs), np.array(ys)

time_steps = 10
X_seq, y_seq = create_sequences(X_scaled, y_scaled, time_steps)

# 5. Split into training and testing sets
split_idx = int(0.8 * len(X_seq))
X_train, X_test = X_seq[:split_idx], X_seq[split_idx:]
y_train, y_test = y_seq[:split_idx], y_seq[split_idx:]

# 6. Build the LSTM model
model = Sequential()
model.add(LSTM(50,                              return_sequences=True,
input_shape=(X_train.shape[1], X_train.shape[2])))
model.add(LSTM(50))
model.add(Dense(1))

model.compile(optimizer='adam', loss='mse')

# 7. Train the model
history      =      model.fit(X_train,      y_train,      epochs=50,
validation_split=0.1, batch_size=32, verbose=1)
from tensorflow.keras.callbacks import EarlyStopping

early_stop     =     EarlyStopping(monitor='val_loss',     patience=5,
restore_best_weights=True)

model.fit(X_train,    y_train,    epochs=50,    validation_split=0.1,
batch_size=32,
          callbacks=[early_stop], verbose=1)
# 8. Predict and evaluate
y_pred = model.predict(X_test)
y_pred_inv = scaler_y.inverse_transform(y_pred)
```

```
y_test_inv = scaler_y.inverse_transform(y_test.reshape(-1, 1))

mae = mean_absolute_error(y_test_inv, y_pred_inv)
rmse = np.sqrt(mean_squared_error(y_test_inv, y_pred_inv))
r2 = r2_score(y_test_inv, y_pred_inv)

print(f"MAE: {mae:.4f}")
print(f"RMSE: {rmse:.4f}")
print(f"R²: {r2:.4f}")

# 9. Plot predictions
plt.figure(figsize=(10, 5))
plt.plot(y_test_inv, label='Actual Loyalty')
plt.plot(y_pred_inv, label='Predicted Loyalty')
plt.title('Actual vs Predicted Loyalty')
plt.xlabel('Time')
plt.ylabel('Loyalty Score')
plt.legend()
plt.grid(True)
plt.show()
plt.savefig('actual_vs_predicted_loyalty.png',                dpi=300,
bbox_inches='tight')
plt.savefig('training_loss_plot', dpi=300, bbox_inches='tight')

# 10. Export results
metrics_df = pd.DataFrame({
    'Metric': ['MAE', 'RMSE', 'R-squared'],
    'Value': [mae, rmse, r2]
})
metrics_df.to_csv('model_evaluation_metrics.csv', index=False)

predictions_df = pd.DataFrame({
    'Actual': y_test_inv.flatten(),
    'Predicted': y_pred_inv.flatten()
})
predictions_df.to_csv('model_predictions.csv', index=False)

model.save('loyalty_lstm_model.h5')
```